IoT Honeypot Project – Data Collection & Initial Analysis

Introduction

As part of my cybersecurity learning, I designed and deployed a honeypot on AWS to observe real-world attack behaviour. The aim of this project was to move beyond theory and gain practical insight into how attackers interact with exposed systems.

The honeypot was intentionally configured to simulate a vulnerable environment, allowing me to safely capture and analyse malicious activity such as login attempts, command execution, and network patterns.

Initial Deployment on AWS

The honeypot was hosted on an AWS virtual machine with publicly exposed services, including SSH and HTTP and 15 other services. This setup was designed to attract automated attacks and simulate a real-world target.

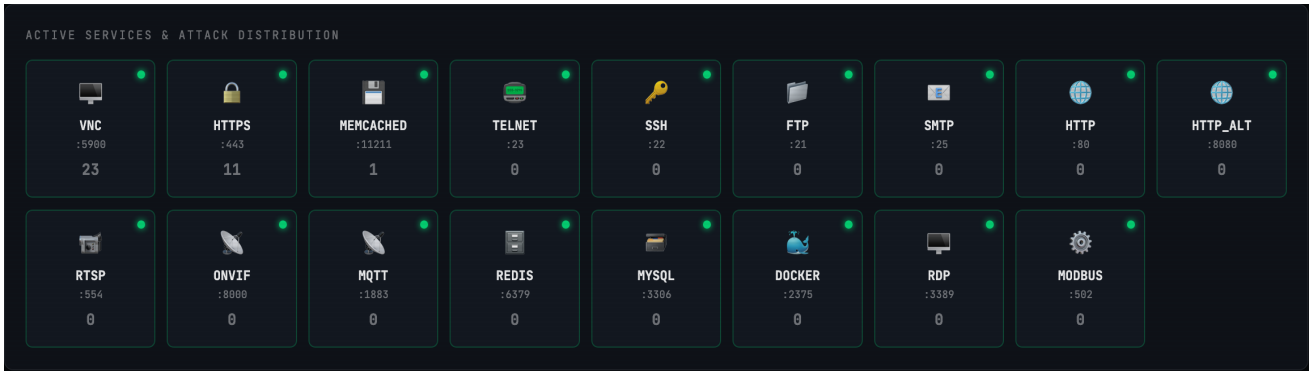
Shortly after deployment, the system began receiving traffic from various sources. The collected data includes:

- Brute-force login attempts
- Requests to HTTP services
- Source IP addresses from different regions
- Automated scripts attempting to interact with the system

Screenshots included below shows examples of these activities and how attackers attempted to gain access.

Screenshots:

1.



These are the services I used in the honeypot. Even though some of them aren't directly related to IoT devices, I included them to see how attacks would work on a broader setup.

2.

The screenshots below provide an overall summary, including total running time, total number of attacks, top attacking countries, most targeted ports, and other key statistics.

honeyPot LIVE

TOTAL ATTACKS	UNIQUE IPS	COUNTRIES	UPTIME
38	20	13	5m

Hikvision DS-2CD2043G2-I | V5.7.15

MONITORING
- Overview 38
- Attack Map
- Live Log

INTELLIGENCE
- Sessions 31
- CVE Exploits 23
- Credentials
- Malware URLs
- Honeytokens

ANALYSIS
- Services
- Report

honeyPot v1.0.0
5m

LIVE FEED — 38 total attacks captured /// 1 CVE exploit attempts detected ///

THREAT INTELLIGENCE REPORT — Last 7d

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honeyPot — Threat Intelligence Report

Generated:    10/03/2026, 22:07:12
Period:       Last 168h
Device:       Hikvision DS-2CD2043G2-I (V5.7.15)

── EXECUTIVE SUMMARY ──

Total Attacks:         19,061
Unique Source IPs:     954
Countries:             58
CVE Exploits:          23
Botnet Credentials:    327
Malware URLs Captured: 0
Honeytokens Triggered: 296

── TOP ATTACKING COUNTRIES ──
1. Ireland  2,690 attacks (14.1%)
2. United States  2,460 attacks (12.9%)
3. Brazil  1,698 attacks (8.9%)
4. Denmark  1,527 attacks (8.0%)
5. Pakistan  1,348 attacks (7.1%)

── TOP ATTACKING IPs ──
1. [redacted]1  1,142 hits  (United States)
2. [redacted]  485 hits  (The Netherlands)
3. [redacted]  388 hits  (United Kingdom)
4. [redacted]7  234 hits  (Estonia)
5. [redacted]  229 hits  (Estonia)

── CVE EXPLOITS ──
1. CVE-2023-44487  [HIGH]  13 attempts  — HTTP/2 Rapid Reset DDoS
2. REDIS-RCE  [CRITICAL]  5 attempts  — Redis CONFIG SET RCE
3. DOCKER-ESCAPE  [CRITICAL]  4 attempts  — Docker API Container Escape
4. CVE-2021-36260  [CRITICAL]  1 attempts  — Hikvision Command Injection

── TOP SERVICES ATTACKED ──
1. vnc  :5900  14,087 hits  (73.9%)
2. http  :80  1,755 hits  (9.2%)
3. telnet  :23  1,689 hits  (8.9%)
4. https  :443  998 hits  (5.2%)
5. onvif  :8000  148 hits  (0.8%)

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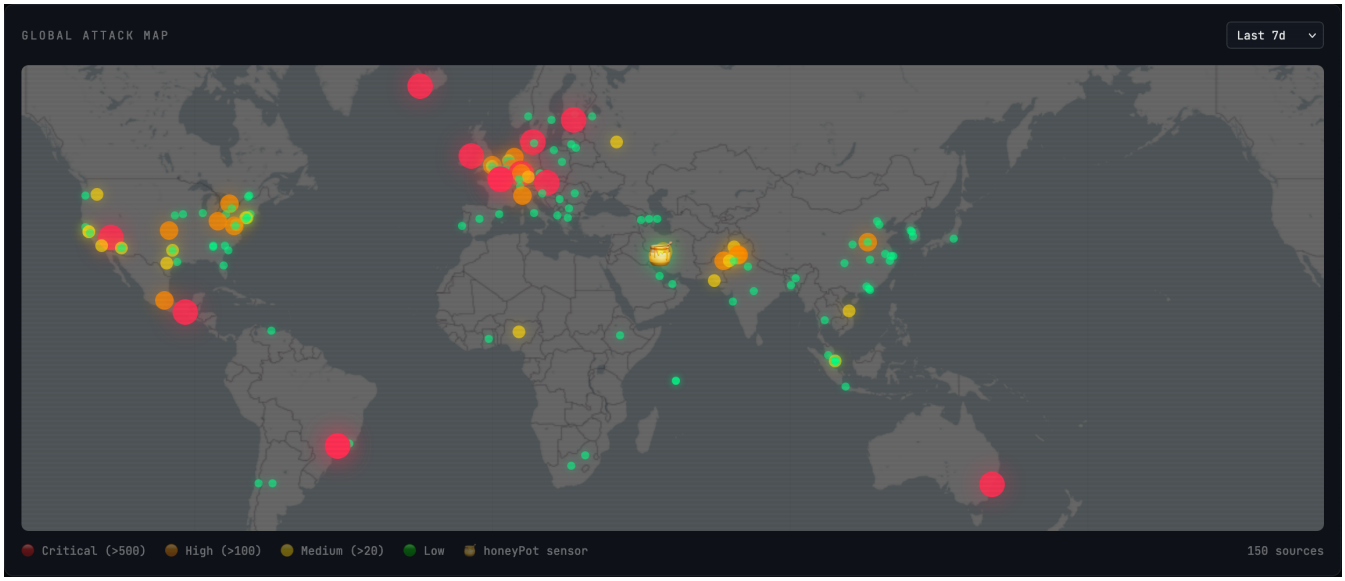
3.

World map showing the origin of attacks: green indicates low activity, orange indicates medium, yellow indicates high, and red indicates critical levels of attacks.

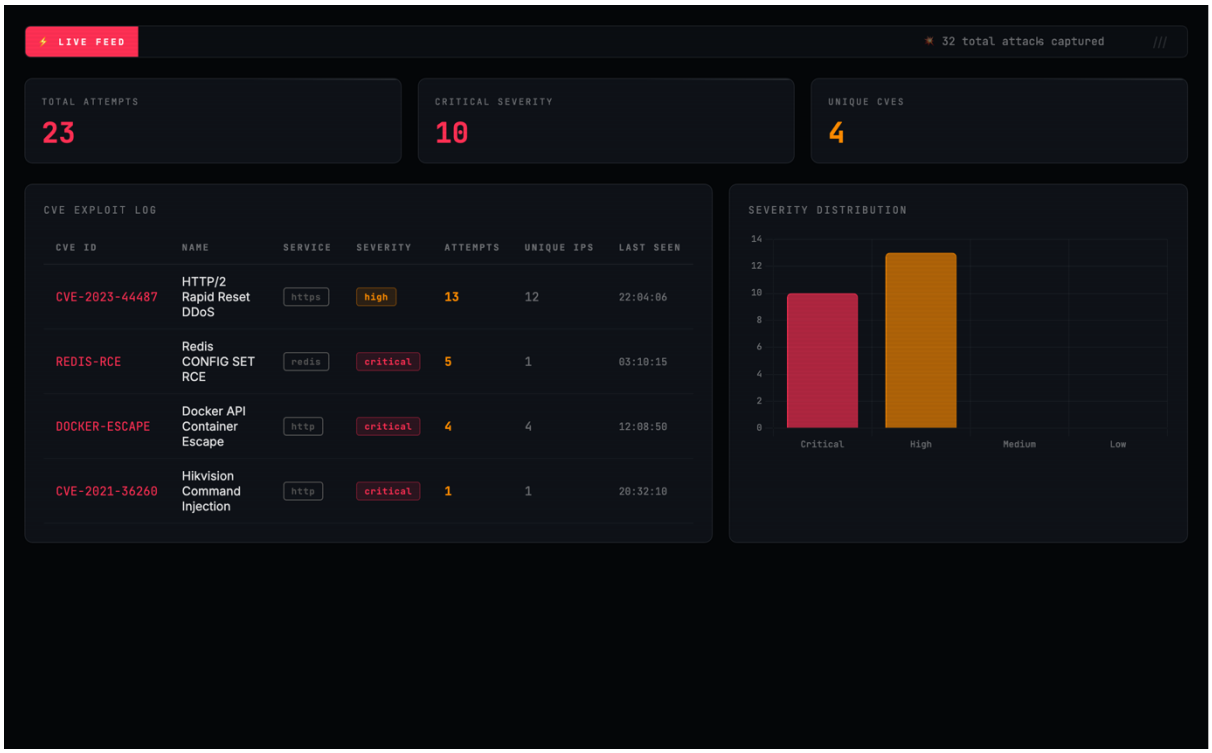


4.

These screenshots show that the highest number of attacks originated from Ireland (2.7k), followed by the United States (2.5k).

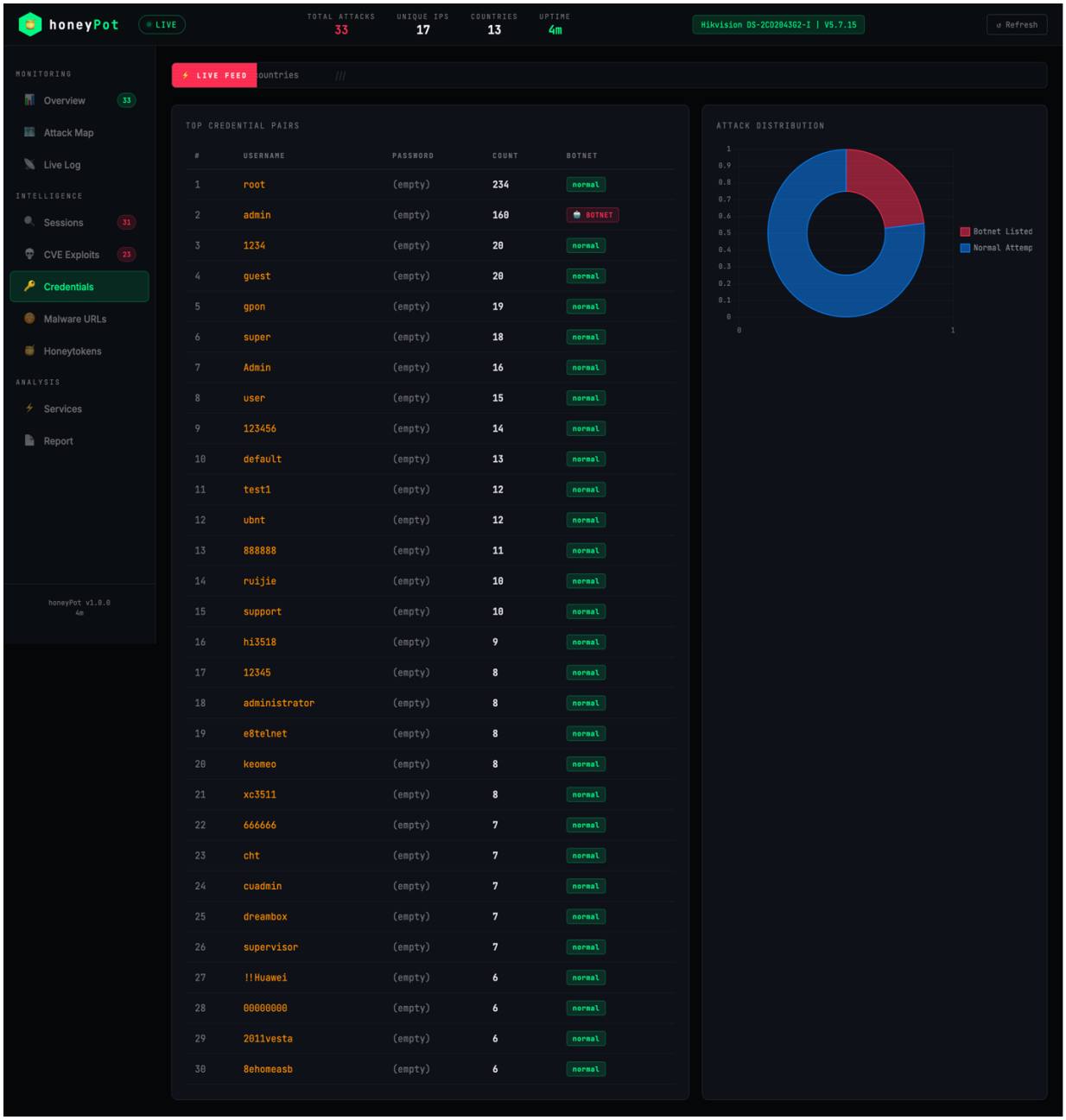
5.

The screenshot below shows the logs of CVE exploits.



6.

The screenshots below show the credentials that were used on our services.



7.

These are the times our honeypot honeytokens were triggered, showing the attacker's country, the targeted service, and the actions they attempted.

honeyPot LIVE

TOTAL ATTACKS	UNIQUE IPS	COUNTRIES	UPTIME
33	17	13	4m

Hikvision DS-2CD2043G2-I | V5.7.15

Refresh

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honeyPot v1.0.0
4m

LIVE FEED — 33 total attacks captured /// 1 CVE explit attempts detected /// Attacks from

TOTAL TRIGGERS	UNIQUE IPS	MOST HIT FILE
296	44	/.env

HONEYTOKEN TRIGGER LOG

TIME	IP	COUNTRY	TYPE	VALUE	SERVICE
14:32:16	[redacted]	The Netherlands	HTTP_GET	/admin/config.php	http
14:32:06	[redacted]	The Netherlands	HTTP_GET	/admin/config.php	http
14:13:45	[redacted]	Poland	HTTP_GET	/.env	http
13:52:38	[redacted]	Ghana	HTTP_GET	/admin/config.php	http
11:34:54	[redacted]	United States	HTTP_GET	/.env	http
09:40:54	[redacted]	United States	HTTP_GET	/.git/config	http
09:32:25	[redacted]	France	HTTP_GET	/.env	http
07:29:39	[redacted]	Albania	HTTP_GET	/.env	http
06:52:15	[redacted]	United States	HTTP_GET	/api/.env	http
06:52:15	[redacted]	United States	HTTP_GET	/config/.env	http
06:12:59	[redacted]	The Netherlands	HTTP_GET	/admin/config.php	http
06:12:47	[redacted]	The Netherlands	HTTP_GET	/admin/config.php	http
06:03:03	[redacted]	United States	HTTP_GET	/.env	http
02:17:00	[redacted]	United States	HTTP_GET	/.env	http
00:59:46	[redacted]	United States	HTTP_GET	/.git/config	http
00:18:46	[redacted]	The Netherlands	HTTP_GET	/admin/config.php	http
22:36:06	[redacted]	Serbia	HTTP_GET	/.env	http
21:17:57	[redacted]	United States	HTTP_GET	/.env	http
19:33:51	[redacted]	The Netherlands	HTTP_GET	/id_rsa	http
19:33:50	[redacted]	The Netherlands	HTTP_GET	/.ssh/id_rsa	http
19:33:50	[redacted]	The Netherlands	HTTP_GET	/.env.local.php	https
19:33:50	[redacted]	The Netherlands	HTTP_GET	/web.config	http
19:33:49	[redacted]	The Netherlands	HTTP_GET	/wp-config.php~	http
19:33:49	[redacted]	The Netherlands	HTTP_GET	/.ssh/id_rsa	https
19:33:49	[redacted]	The Netherlands	HTTP_GET	/web.config	https
19:33:49	[redacted]	The Netherlands	HTTP_GET	/wp-config.php.old	http
19:33:49	[redacted]	The Netherlands	HTTP_GET	/wp-config.php.swp	https
19:33:48	[redacted]	The Netherlands	HTTP_GET	/docker-compose.yml	http
19:33:47	[redacted]	The Netherlands	HTTP_GET	/docker-compose.yml	https
19:33:47	[redacted]	The Netherlands	HTTP_GET	/wp-config.php.save	http

BY FILE / CREDENTIAL

1	/.env	46
2	/admin/config.php	11
3	/.git/config	11
4	/.env.development.local	9
5	/wp-config.php.old	8
6	/v1/.env	8
7	/api/.env	7
8	/config/.env	7
9	/wp-config.php.bak	7
10	/config.php	7
11	/admin/.env	7
12	/.env.local	7
13	/.ssh/id_rsa	6
14	/wp-config.php~	6
15	/wp-config.php.save	6
16	/private/.env	6
17	/.aws/credentials	6
18	/src/.env	6
19	/v2/.env	6
20	/.env.prod	6
21	/.env.production	6
22	/wp-config.php.swp	5
23	/docker-compose.yml	5
24	/core/.env	5
25	/.env.bak	5
26	/.env~	5
27	/web/.env	5
28	/laravel/.env	5
29	/.env.old	5
30	/.env.production.local	5
31	/.env.docker	5
32	/.env.staging	5
33	/wp-config.php	4
34	/backend/.env	4
35	/.env.backup	4
36	/server/.env	4
37	/.env.dev	4
38	/.env.example	4
39	/.env.development	4
40	/app/.env	4
41	/.env.local.php	3
42	/web.config	3
43	/.env.swp	3
44	/id_rsa	2
45	/wp-config.php.orig	2
46	/public/.env	1
47	/application.config.php	1
48	/test.config.php	1
49	/aws_config.php	1
50	/setup-config.php	1
51	/aws_secret_config.php	1
52	/wp-admin/setup-config.php	1

First Results

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The initial results showed that the system was targeted almost immediately after being exposed to the internet.

Key observations:

- High volume of attacks received on VNC port, following by HTTP, Telnet, HTTPS and ONVIF.
- Repeated attempts from multiple IP addresses, indicating automated attacks
- Basic interaction with HTTP endpoints

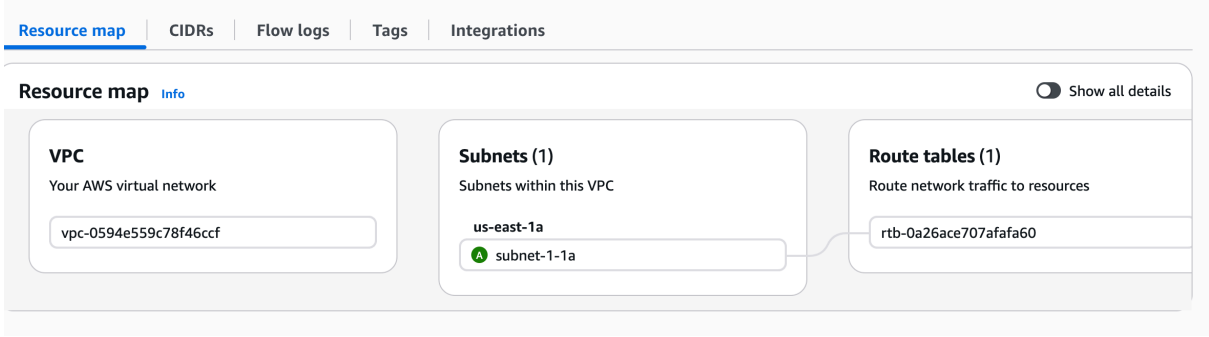
These findings highlight how quickly exposed systems are discovered and targeted.

Supervisor Feedback:

1. Fix the SSH port configuration.
2. Make the HTTP service more realistic.
3. Add VPN/Tor endpoints to track attacks coming through them.
4. Satisfied with the dashboard setup.

My Observation:

1. Our honeypot was hosted in the **US-East-1** subnet, and most of the attacks originated from this region.



2. Our IP wasn't showing on the website like Shodan and Censys.
3. Services weren't realistic enough
4. There was lack of features in dashboard such as filtering data, where are the tools attackers using to attack, how they find our honeypot.